

Primary Exposition Gap Register

Claims removed pending exact Lean counterparts

A research-frontier companion to *The Fabius Function and Rvachev's Up-Function*
based on Vladimir Reshetnikov's `ProveIt` formal development

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Abstract

This notebook preserves mathematical material removed from the primary Fabius exposition during its proof-correspondence audit. Nothing here is claimed as a theorem of the Lean development unless explicitly described as formal background. Each item needs a new public theorem, a precise cost or convergence model, or a correction to its statement. The larger endpoint-asymptotic frontier is maintained separately in the sibling notebook *Small-Argument Asymptotics*.

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1 Status and promotion rule

The primary exposition admits a mathematical statement only when an actual proved Lean declaration has the same hypotheses and conclusion. An elementary consequence, a paper proof, a source comment, an executable calculation, or an equivalent normalization is not sufficient. Promotion from this register requires a public Lean theorem, a successful build, an exact source citation in the exposition, and a rebuilt, visually checked PDF.

2 Fourier-decay normalization

The source proves polynomially weighted decay. The conventional shifted-weight form that was removed is:

Candidate statement 2.1. For every $N \in \mathbb{N}$, there is $C_N > 0$ such that

$$|\widehat{\text{up}}(\xi)| \leq C_N(1 + |\xi|)^{-N} \quad (\xi \in \mathbb{R}).$$

It is standardly equivalent to the proved estimate after a bounded-region argument and a change of constant, but it has no literal named Lean bridge.

Formalization obligation 2.2. Derive the shifted-weight estimate, including $|\xi| \leq 1$, and expose it as a public declaration.

3 Dyadic specializations and evaluator cost

3.1 Concrete rational values

Let F denote the bounded Fabius function and up Rvachev’s compactly supported up-function. The notation $c_n = \int_{-1}^1 x^{2n} \text{up}(x) \, dx$ is the even-moment notation of the primary article. The general dyadic evaluator is proved correct, but these specializations have no named theorems:

Candidate statement 3.1.

$$c_1 = \frac{1}{9}, \quad F(1/4) = \frac{5}{72}, \quad F(1/8) = \frac{1}{288}, \quad F(3/8) = \frac{73}{288}.$$

Formalization obligation 3.2. Prove named specializations by reducing the exact evaluator and bridge each result to the analytic function.

3.2 Popcount complexity

Termination is formalized, but the removed prose did not distinguish the following two possible complexity claims. For $e, a \in \mathbb{N}$ with $0 < a < 2^e$, let $D(e, a)$ be the number of recursive calls that enter the evaluator’s interior branch.

Candidate statement 3.3. The exact-count candidate is $D(e, a) = \text{wt}_2(a)$.

Candidate statement 3.4. The weaker upper-bound candidate is $D(e, a) \leq \text{wt}_2(a)$.

Formalization obligation 3.5. Define D in Lean, settle which statement matches the executable recursion, and prove the resulting relation to binary weight, including endpoint and saturation branches.

3.3 Independent analytic derivations

Earlier prose derived the dyadic formula through Taylor integrals, block substitutions, and the exact remainder $-F(y)$. The final rational identities are formal, but those intermediate analytic equations are not public theorem-for-theorem counterparts.

Formalization obligation 3.6. Formalize named Taylor integral identities for the bounded and signed global functions, including the blockwise changes of variables and the exact remainder.

4 Complex-shift spline estimate

The proved estimate has the factor $\exp(|Q - \frac{1}{2}|)$. Put

$$N_p(x) = \max\left(0, \left\lfloor 2^p x + \frac{1}{2} \right\rfloor\right), \quad S_p^{(Q)}(x) = \frac{(-1)^p}{2^{\binom{p}{2}} p!} \sum_{r=0}^{N_p(x)-1} (-1)^{\text{wt}_2(r)} (r - 2^p x + Q)^p,$$

with an empty sum when $N_p(x) = 0$. The removed proposed sharpening was:

Candidate statement 4.1. For every $p \in \mathbb{N}$ with $p \geq 1$, every $Q \in \mathbb{C}$, and every $x \in \mathbb{R}$,

$$\left| S_p^{(Q)}(x) - S_p^{(1/2)}(x) \right| \leq 2^{-(p-1)} \left(\exp\left(\left|Q - \frac{1}{2}\right|\right) - 1 \right).$$

Warning 4.2. The accompanying assertion that the relevant triangular exponent attains equality *exactly in the top term* is false when $p = 2$: both endpoint indices attain the same exponent.

Formalization obligation 4.3. Formalize the finite Taylor majorant, state the minimizer edge cases correctly, and prove a public $\exp - 1$ estimate if it survives that correction.

5 Prefix-kernel interpretations

The binomial kernel and the resulting prefix-sum formula are formalized. These two interpretations are not:

- Candidate statement 5.1.*
- for $i, k \in \mathbb{N}$ with $k \geq 1$, $\binom{i+k-1}{k-1}$ counts ordered k -tuples of nonnegative integers summing to i ;
 - for $k \in \mathbb{N}$ with $k \geq 1$, the kernel is the k -fold discrete convolution power of the constant kernel.

Formalization obligation 5.2. Choose a finite-support or locally finite convolution model, prove the counting bijection, and connect it to the existing `iteratedPrefixKernel`.

6 Block-polynomial derivatives at one

The finite factorization and its power-sum consequences are formal. The literal derivative version removed from the article was:

Candidate statement 6.1. For $m, r \in \mathbb{N}$ with $m < r$, the m -th derivative of $\prod_{j < r} (1 - z^{2^j})$ vanishes at $z = 1$, while its r -th derivative there is

$$(-1)^r r! 2^{\binom{r}{2}}.$$

Formalization obligation 6.2. Formalize polynomial iterated derivatives at $z = 1$ and identify them with the proved coefficient or power-sum formula.

7 An actual infinite formal product

Lean proves coefficient stabilization of the finite products $\prod_{j < r} (1 - z^{2^j})$. The stronger construction formerly asserted was:

Candidate statement 7.1. Construct $\prod_{j \geq 0} (1 - z^{2^j})$ as a formal infinite product and prove that it equals the Thue–Morse power series.

Formalization obligation 7.2. Select an appropriate formal-power-series topology or summability API, define the product, and prove equality with the coefficient-stabilized series. Until then, the notation is only shorthand for finite stabilization.

8 Formal-background crosswalk

The following declarations support the proved background cited above. None of them is a counterpart for the adjacent candidate statement.

Background	Lean declaration and module
Polynomially weighted Fourier decay	<code>rvachevFourier_real_rapidDecay</code> in <code>PoissonSummation.lean</code>
Even moments and exact dyadic evaluation	<code>moment</code> , <code>fabiusDyadicValue</code> , <code>fabiusDyadicValue_cast</code> , <code>fabiusDyadic_cast</code> in <code>Arithmetic.lean</code> , <code>PaperStatements.lean</code> , and <code>DyadicAnalytic.lean</code>
Complex-shift exponential bound	<code>norm_fabiusComplexShiftSpline_sub_center_1e_half_pow_mul_exp</code> in <code>FabiusComplexShiftSpline.lean</code>
Prefix kernel and convolution formula	<code>iteratedPrefixKernel</code> and <code>iteratedPrefix_eq_sum_kernel</code> in <code>ThueMorsePrefix.lean</code>
Even/odd Thue–Morse recurrences	<code>binaryWeight_two_mul</code> and <code>binaryWeight_two_mul_add_one</code> in <code>DyadicClosedForm.lean</code>
Finite block product and coefficient stabilization	<code>thueMorseBlockPolynomial_eq_product</code> and <code>coeff_finite_thueMorse_product</code> in <code>ThueMorseGenerating.lean</code>

9 Endpoint material maintained elsewhere

The sibling notebook *Small-Argument Asymptotics* preserves the removed raw Mellin-inversion derivation, exponential Gamma–zeta decay estimates, numerical periodic profiles, proposed ordered-composition formulas for saddle coefficients, explicit higher Lambert polynomials, a proposed leading-derivative law, uniqueness and phase-density arguments, half-moment asymptotics, and strengthened endpoint comparisons.

The primary article now retains only the exact negative-Laplace decomposition, the proved Gamma–zeta Fourier series, the evaluated constant, lower-Lambert reduction, Bromwich inversion, the first two periodic saddle corrections, and the recursive all-orders theorem.

10 Provenance

This register was created from the claim-by-claim audit of the former drafts and primary exposition on 25 August 2026. Related numerical and exploratory material is also preserved in the dyadic, q -binomial, inverse, integration, and repeated-integration notebooks in this research-frontier directory.